

The Transport Layer: TCP and UDP

TCP (Transmission Control Protocol):

- **Connection-Oriented:** A "handshake" is required before data is sent.
- **Reliable:** If a packet is lost, TCP detects it and resends it.
- **Ordering:** Packets arrive in the exact order they were sent.
- **Use Case:** Web browsing (HTTP), Email (SMTP), and File Transfers (FTP).

UDP (User Datagram Protocol):

- **Connectionless:** No handshake; just sends data immediately.
- **Unreliable:** "Fire and forget." If a packet is lost, it is gone forever.
- **Speed:** Much faster than TCP because there is no overhead for checking errors.
- **Use Case:** Video streaming, Online gaming, and Voice over IP (VoIP).